

Adding Three-Digit Numbers with Regrouping

Answer Key

Grades 2–3 Number and Operations in Base Ten

Quick Answers

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|--------|-------------|-------------|---------|
| 1. 473 | 4. 900, 843 | 7. 425 | 10. 733 |
| 2. 438 | 5. 636 | 8. 1024 | |
| 3. 734 | 6. 804 | 9. 900, 901 | |
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Curriculum

Level: Grades 2–3 Number and Operations in Base Ten

2.NBT.B.5 / 3.NBT.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 12 tagged checks.

Common wrong answers

3. *If they got 634: regrouped the ones but dropped the regrouped ten going into the hundreds*
 6. *If they got 794: never carried the regrouped ten into the tens column*
 8. *If they got 914: wrote only the last digit of each column total and carried nothing*
 10. *If they got 623: wrote only the last digit of each column total and carried nothing*
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1. Add $348 + 125$.

Step 1. Ones: $8 + 5 = 13$. That is one ten and three ones, so write 3 under the ones and carry the ten.

Step 2. Tens: $4 + 2 = 6$, plus the carried ten makes 7 tens.

Step 3. Hundreds: $3 + 1 = 4$ hundreds, with nothing to carry.

473

Listen for: a child who writes 13 under the ones column. Only one digit belongs in each column.

2. Add $265 + 173$.

Step 1. Ones: $5 + 3 = 8$. Under ten, so nothing is carried.

Step 2. Tens: $6 + 7 = 13$ tens. Write 3 in the tens column and carry one hundred.

Step 3. Hundreds: $2 + 1 = 3$, plus the carried hundred makes 4.

438

Teaching note: the regroup happens in the tens here, not the ones. Children who have only met ones-column regrouping often miss it.

3. Add $476 + 258$.

Step 1. Ones: $6 + 8 = 14$. Write 4, carry one ten.

Step 2. Tens: $7 + 5 = 12$, plus the carried ten is 13 tens. Write 3, carry one hundred.

Step 3. Hundreds: $4 + 2 = 6$, plus the carried hundred is 7.

734

Common wrong answer: 634 — the first carry was used and the second one forgotten. Two regroupings in a row is where that slip lives.

4. Estimate $487 + 356$, then add exactly.

Step 1. Round each addend to the nearest hundred. 487 is past the halfway mark of 450, so it rounds up to 500; 356 is past 350, so it rounds up to 400.

Step 2. Add the rounded numbers: five hundreds plus four hundreds is nine hundreds.

900

Step 3. Now the exact sum. Ones: $7 + 6 = 13$, write 3 carry one. Tens: $8 + 5 = 13$, plus the carry is 14 tens — write 4, carry one hundred. Hundreds: $4 + 3 = 7$, plus the carry is 8.

843

Compare: both addends rounded up, so the estimate sits above the true sum. An answer near nine hundred is what we wanted to see.

5. Add $389 + 247$.

Step 1. Ones: $9 + 7 = 16$. Write 6, carry one ten.

Step 2. Tens: $8 + 4 = 12$, plus the carried ten is 13 tens. Write 3, carry one hundred.

Step 3. Hundreds: $3 + 2 = 5$, plus the carried hundred is 6.

636

Check it: 389 is nearly 400 and 247 is nearly 250, so a sum just above 600 is right where it should be.

6. Add $596 + 208$.

Step 1. Ones: $6 + 8 = 14$. Write 4, carry one ten.

Step 2. Tens: $9 + 0 = 9$, but the carried ten still has to go in, so the column is 10 tens. Write 0 in the tens and carry one hundred.

Step 3. Hundreds: $5 + 2 = 7$, plus the carried hundred is 8.

804

Common wrong answer: 794 — the carry out of the ones was never added to the tens. A zero in a column does not mean the column is finished.

7. Maya's class: 268 cans in October and 157 in November.

Step 1. The question asks for the two collections put together, and putting amounts together is addition: $268 + 157$.

Step 2. Ones: $8 + 7 = 15$. Write 5, carry one ten. Tens: $6 + 5 = 11$, plus the carry is 12 tens — write 2, carry one hundred.

Step 3. Hundreds: $2 + 1 = 3$, plus the carried hundred is 4.

425

Check it: about 270 and about 160 together is about 430, so this answer is the right size.

8. Add $675 + 349$.

Step 1. Ones: $5 + 9 = 14$. Write 4, carry one ten.

Step 2. Tens: $7 + 4 = 11$, plus the carried ten is 12 tens. Write 2, carry one hundred.

Step 3. Hundreds: $6 + 3 = 9$, plus the carried hundred is 10 hundreds — which is one thousand. Write 0 in the hundreds place and 1 in a new thousands place.

1024

Common wrong answer: 914 — only the last digit of each column was written down and no carry was used anywhere.

9. Estimate $612 + 289$, then add exactly.

Step 1. Round to the nearest hundred: 612 is only a little past 600, so it rounds down to 600; 289 is close to 300, so it rounds up.

Step 2. Six hundreds plus three hundreds is nine hundreds.

900

Step 3. Exact sum. Ones: $2 + 9 = 11$, write 1 carry one ten. Tens: $1 + 8 = 9$, plus the carry is 10 tens — write 0, carry one hundred. Hundreds: $6 + 2 = 8$, plus the carry is 9.

901

Why so close: one addend rounded down by twelve and the other rounded up by eleven, so the two errors almost cancelled. That is the estimate working at its best, and it is worth pointing out.

10. Ben wrote $457 + 276 = 623$. Add it correctly.

Step 1. Ben's columns were right but his carries were missing: the ones came to 13 and the tens to 12, and he kept only the last digit of each without passing the ten along.

Step 2. Done properly: ones $7 + 6 = 13$, write 3 and carry one ten. Tens $5 + 7 = 12$, plus the carried ten is 13 tens — write 3, carry one hundred.

Step 3. Hundreds: $4 + 2 = 6$, plus the carried hundred is 7.

733

Common wrong answer: 623 — exactly Ben's. It is more than one hundred short of the truth, which is what a lost carry costs.